

Answer: C

39 From graph,

$$900 = 1000e^{-\lambda_x \cdot 4} \implies e^{4\lambda_x} = \frac{10}{9} \implies 4\lambda_x = \ln\left(\frac{10}{9}\right) \implies \lambda_x = \frac{1}{4} \ln\left(\frac{10}{9}\right)$$

$$900 = 1000e^{-\lambda_y \cdot 2} \implies e^{2\lambda_y} = \frac{10}{9} \implies 2\lambda_y = \ln\left(\frac{10}{9}\right) \implies \lambda_y = \frac{1}{2} \ln\left(\frac{10}{9}\right)$$

At $t = 8$ h,

$$A_x = 1000e^{-\lambda_x \cdot 8} = 1000e^{-\frac{1}{4} \ln\left(\frac{10}{9}\right) \cdot 8} = 1000e^{-2 \ln\left(\frac{10}{9}\right)} = 1000e^{-2 \ln\left(\frac{10}{9}\right)}$$

$$A_y = 1000e^{-\lambda_y \cdot 8} = 1000e^{-\frac{1}{2} \ln\left(\frac{10}{9}\right) \cdot 8} = 1000e^{-4 \ln\left(\frac{10}{9}\right)} = 1000e^{-4 \ln\left(\frac{10}{9}\right)}$$

$$\frac{A_x}{A_y} = e^{2 \ln\left(\frac{10}{9}\right)} = e^{\ln\left(\frac{100}{81}\right)} = \frac{100}{81}$$

Answer: D

40 Proton number of Y = 93